Short Answer Test

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1. Define cloud computing and Explain any four characteristics that make it different from traditional on-premise computing

2. Trace the evolution of cloud computing from hardware evolution to Internet software evolution and show how these changes enabled modern cloud platforms

3. Explain the role of server virtualisation in cloud computing. Differentiate btw virtualisation as an enabling technology and Cloud computing as a service model

4. Discuss the role of data centers in delivering cloud services

5. Differentiate Iaas, Paas, Saas and Xaas with one suitable real world suitable example.

ANSWERS

1. Definition of cloud computing;

* Cloud computing is a technology that provides computing resources such as servers, storage, databases, networking and software over the Internet whenever they are needed

Four characteristics

1. on demand self service
2. Broad network access
3. Resource pooling
4. Rapid elasticity

1. on demand self service :

users can get computing resources whenever they require

2. Broad network access

Cloud services can be accessed through the internet

3. Resource pooling

Cloud providers share their large pool of computing resources

4. Rapid Elasticity :

Resources can be increased (or) decreased quickly

2.

Evolution :

1. Hardware Evolution :

* computers were large and Expensive

⇒ Minicomputers → personal computers → powerful servers

* Hardware became smaller, faster and cheaper

2. Networking Evolution :

Computers were connected through networks

The development of the internet allowed computers to communicate globally

3. Virtualisation :

Physical servers could be divided into multiple virtual Machines

This improved hardware utilization and reduced cost

4. Internet software Evolution :

Web applications became more powerful

Applications can be accesed through browser instead of being installed locally.

5. Web 2.0 and service oriented Applications :

Applications became interactive and service based

users could access services from anywhere through the Internet.

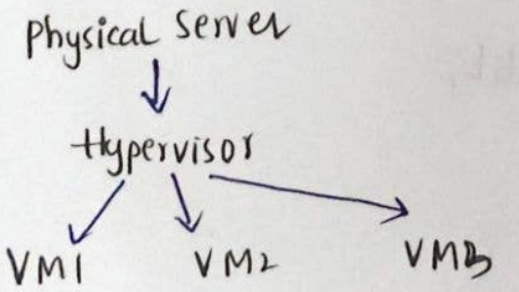
3.

Role of server virtualisation in cloud computing :

Server Virtualisation :

It is a process of dividing one physical server into multiple virtual serves using software called as hypervisor

Example :



Role in cloud computing :

1. Better resource utilization :

   Multiple VMs can use the same physical server

2. Scalability :

   New virtual machines can be created quickly

3. Cost reduction :

   Few physical servers are required

4. Isolation :

   Different users can run their applications

5. Easy Management :

   VMs can be created, moved copied, deleted easily

4.

Role of Data centers in cloud computing

1. stores data
2. provides servers
3. provides networking
4. provides security
5. Ensures availability
6. Supports scalability

5.

Iaas - (Infrastructure as a service)

* provides severs, storage and networking
* Ex: Amazon EC2

Paas (platform as a service)

* provides a platform to develop and run application
* Ex: Google App Engine

Saas (software as a service)

* provides ready to use software
* Ex: Microsoft 365

Xaas (Anything as a service)

* provides almost any IT service through internet
* AWS lamda → Example